1D Seismic Trace Visualizer

Forward Modeling Architecture for Subsurface Exploration

VED DIPAKKUMAR PARMAR

Project Objective

- Simulate computational seismic surveys.
- Generate 1D synthetic wiggle traces.
- Link well logs to surface data.



Geophysical Foundation

Source Wavelet

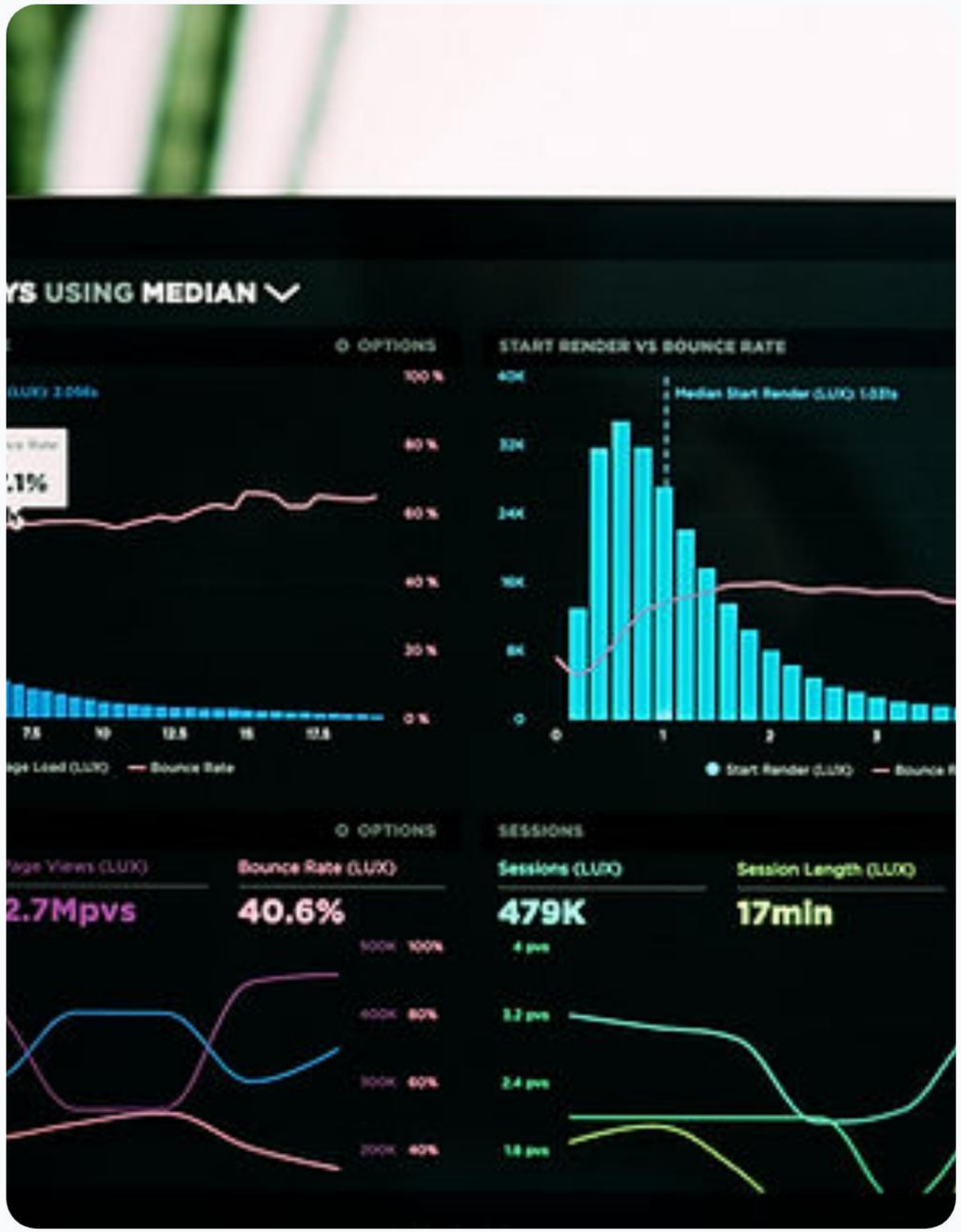
30 Hz Ricker wavelet energy.

Earth Reflectivity

Reflection Coefficients map boundaries.

Convolution

Combines wavelet and earth response.



Software Architecture

Core Engine

High-performance NumPy array math.

Visualization

Industry-standard Matplotlib trace rendering.



Simulation Complete

Translating theoretical physics into practical data-processing workflows.

Image Sources



https://images.stockcake.com/public/5/5/b/55ba31f1-aab7-4ccb-abe1-4fb3332e156e_large/layered-earth-textures-stockcake.jpg

Source: stockcake.com



https://static.wixstatic.com/media/nsplsh_5e1c2facf7da4793b9b68b20a019326f~mv2.jpg/v1/fill/w_568,h_378,al_c,q_80,usm_0.66_1.00_0.01,enc_avif,quality_auto/nsplsh_5e1c2facf7da4793b9b68b20a019326f~mv2.jpg

Source: www.insightbig.com